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# RoboMME: Benchmarking and Understanding Memory for Robotic Generalist Policies

*16 Tasks Across Four Memory Categories Reveal That Current VLA Memory Mechanisms Are Far From Solving Long-Horizon Manipulation*

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Vision-language-action models have made remarkable progress on short-horizon manipulation, but long-horizon tasks demanding memory remain poorly understood. RoboMME introduces the first large-scale standardized benchmark for memory in robotic generalist policies, comprising 16 tasks across temporal, spatial, object, and procedural memory categories, and a systematic study of 14 memory-augmented VLA variants. The result: every variant fails significantly on tasks requiring 5+ step sequences, with the oracle gap exceeding 45%.

## AT A GLANCE

**Problem:** VLA evaluations focus on memoryless tasks; no standardized benchmark assesses how well different memory architectures support long-horizon, history-dependent manipulation.

**Why it matters:** Real-world manipulation involves counting actions, retrieving occluded objects, and executing multi-step procedures — all fundamentally memory-dependent.

**Method:** 16 tasks classified into four memory types; 14 VLA variants built on the  $\pi_{0.5}$  backbone systematically covering representation and integration strategy choices.

**Result:** FrameSamp+Modul achieves best aggregate performance; all variants fail on 5+ step procedural sequences.

**Evidence:** Evaluated on LIBERO and LIBERO-Plus with procedurally generated variants ensuring evaluation diversity.

## The Big Picture

VLA models extend multimodal LLMs with action outputs, achieving impressive generalization across robot platforms and tasks. Yet every established benchmark

— LIBERO, SimplerEnv, MetaWorld — evaluates performance on tasks solvable from the current observation alone, without reference to history. This memoryless evaluation regime is convenient but incomplete.

Real-world manipulation is deeply history-dependent. A robot sorting items must remember which bin each object was assigned to. A robot following a recipe must track which ingredients have already been added. A robot retrieving objects from a drawer must remember where the object was before it was occluded. Without a principled benchmark for these capabilities, the field has no way to measure progress on long-horizon robustness.

RoboMME fills this gap.

## Why Existing Approaches Fall Short

**Benchmark limitations:** Existing benchmarks either test single tasks independently (no temporal dependency) or use cumulative success rate on fixed-length sequences, conflating task difficulty with memory requirement. RoboMME’s taxonomy separates these confounds.

**Architecture comparisons:** Prior work comparing memory mechanisms (frame stacking, recurrent states, external memory) uses different tasks, backbones, and training procedures, making conclusions non-transferable. RoboMME fixes the backbone ( $\pi_{0.5}$ ) and training recipe, varying only the memory component.

## Core Mechanism

### Memory taxonomy (four types):

- **Temporal memory:** Track a sequence of subgoals over time (e.g., place five objects in a specified order; count repeated actions).
- **Spatial memory:** Remember and retrieve the location of objects that become temporarily occluded during the episode.
- **Object memory:** Identify and track specific objects across visual changes (different lighting, partial occlusion, displacement).
- **Procedural memory:** Execute a multi-step procedure from an instruction, where the full procedure is not visible in the current observation.

**Memory mechanism variants (14):** Each variant combines a *representation* (FrameSamp with  $k \in \{1, 4, 8\}$  frames; recurrent hidden state; explicit key-value external memory) with an *integration strategy* (early concatenation; modulation of the VLM trunk; late fusion into the action expert). The full factorial design yields 14 meaningful combinations.

## Technical Formulation

**Memory score:** For task  $\tau$  and memory category  $c$ ,

$$S_\tau = \frac{1}{N_\tau} \sum_{i=1}^{N_\tau} \mathbf{1}[\text{episode } i \text{ succeeds}], \quad M_c = \frac{1}{|T_c|} \sum_{\tau \in T_c} S_\tau.$$

**Memory dependence gap:** For each task  $\tau$ ,

$$\Delta_\tau = S_\tau(\text{memory agent}) - S_\tau(\text{Markov agent}),$$

where the Markov agent uses only the current observation (history length 1). Tasks with  $\Delta_\tau > 0.1$  are classified as genuinely memory-dependent.

**Oracle gap:** The gap to an oracle with perfect recall of all history,

$$\Delta_\tau^{\text{oracle}} = S_\tau(\text{oracle}) - \max_{\text{variant}} S_\tau(\text{variant}).$$

For procedural memory tasks,  $\Delta_\tau^{\text{oracle}} \approx 0.45$  on average — current mechanisms capture at most 55% of the available memory signal.

**FrameSamp+Modul** cross-attends  $k$  sampled historical frames into the VLM backbone via:

$$\mathbf{h}_t = \text{VLM}(o_t) + \text{CrossAttn}(\text{VLM}(o_t), \{h_{t_j}\}_{j=1}^k),$$

where  $\{t_j\}$  are selected historical timesteps.

## Learning or Inference Procedure

All 14 variants share the same training recipe:

1. Pre-train on heterogeneous robot demonstration data (OXE-Mix).
2. Fine-tune on RoboMME task demonstrations with per-variant memory module.
3. Evaluate on held-out procedurally generated episodes (100 per task).
4. Report  $S_\tau$  and  $\Delta_\tau$  for all tasks;  $M_c$  per category.

## What the Guarantee Says

RoboMME does not provide formal guarantees, but establishes two empirical claims with high statistical confidence ( $p < 0.01$  across all reported gaps): (1) 12 of 16 tasks show  $\Delta_\tau > 0.1$ , confirming genuine memory dependence. (2) No variant achieves  $S_\tau > 0.6$  on tasks requiring 5+ step procedural sequences.

## Experimental Findings

- **FrameSamp+Modul:** Best aggregate  $M_c$  across all four categories; particularly strong on temporal counting and time-sensitive tasks.
- **Recurrent:** Best on procedural memory tasks (leverages hidden state compression), but degrades on temporal counting.

- **ExternalMem:** Marginal gains over FrameSamp on most tasks; naive key-value memory is not yet a silver bullet.
- **Dense history (FrameSamp  $k = 8$ ):** Surprisingly weak; long raw history contexts confuse the VLM backbone without selective attention.
- **All variants:** Fail significantly on 5+ step procedural sequences, with oracle gap  $> 45\%$ .

## Ablations and Interpretation

**History length  $k$ :** Performance peaks at  $k = 4$  for most tasks;  $k = 8$  hurts due to context confusion. Task-adaptive  $k$  selection (based on estimated episode length) is a promising direction.

**Integration strategy:** Modulation consistently outperforms early concatenation and late fusion, suggesting that injecting historical context into intermediate VLM representations is more effective than prepending raw tokens or appending to the action expert.

**What memory helps most:** Spatial and object memory benefit most from frame sampling; procedural memory benefits most from recurrence. A hybrid design adapting the memory mechanism to the inferred task type is a natural future direction.

## Reference

Yinpei Dai, Hongze Fu, Jayjun Lee, Yuejiang Liu, Hao-ran Zhang, Jianing Yang, Chelsea Finn, Nima Fazeli, Joyce Chai. **RoboMME: Benchmarking and Understanding Memory for Robotic Generalist Policies.** arXiv:2603.04639, March 2026. <https://arxiv.org/abs/2603.04639>